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Author(s): Sarah Kaplan

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COGNITION, CAPABILITIES, AND INCENTIVES: ASSESSING FIRM RESPONSE TO THE FIBER-OPTIC REVOLUTION

SARAH KAPLAN
University of Pennsylvania

This study examines how CEO cognition, organizational capabilities, and organizational incentives interacted to shape firm strategy during the fiber-optic revolution. I tested for these factors' association with subsequent investment in optical technologies using longitudinal data from 71 communications firms. Results show that each is separately important in shaping outcomes, and their alignment toward the adoption of a new technology leads to the greatest levels of change. In addition, cognition can compensate when organization-level factors are lacking. Considering cognition, capabilities, and incentives together contributes to a more contingent view of the conditions under which CEO cognition matters for firm strategy.

In research on firm response to technical change, a rich literature has established the importance of CEOs and other senior managers in setting strategy during such periods of uncertainty (Eisenhardt & Bourgeois, 1988; Tushman & Rosenkopf, 1996; Virany, Tushman, & Romanelli, 1992). Increasingly, scholars have focused on the specific role of these managers' cognitions (Barr, 1998; Barr, Stimpert, & Huff, 1992; Cho & Hambrick, 2006; Fiol, 1990; Tripsas & Gavetti, 2000). This research has its foundation in a view that strategic action is shaped by how managers notice and interpret change and translate those perspectives into strategic choice (Daft & Weick, 1984). Such a model of organizational action is particularly salient in periods of technical change, when high levels of uncertainty demand that managers' cognitions become the basis for action (Walsh, 1995). Because new technologies are inherently "equivocal" (Weick, 1990: 2), they are subject to "sensemaking" as managers make choices about how to respond. Specifically, Ocasio (1997) and Cho and Hambrick (2006) have suggested that, given the overwhelming amount of (often conflicting) information available during periods of change, what managers pay attention to should affect a firm's strategic direction.

Although these studies have provided compelling

arguments that managerial cognition matters, they have not systematically taken into account how cognition relates to the effects of the organization-level explanations that predominate in the literature on technical change, those of incentives and capabilities. On the one hand, economists have argued that firm investment in new technologies is connected to firm-level incentives (Cockburn & Henderson, 1994; Gilbert & Newbery, 1982; Reinganum, 1983). Specifically, a firm may respond to a new technology if such investments are required to protect a core market. On the other hand, population ecologists and evolutionary theorists have shown that capabilities are an essential driver of firm response to technical change (Hill & Rothaermel, 2003; Tushman & Anderson, 1986), either impeding change because of the inertial forces of existing routines in the current business (Barnett & Carroll, 1995; Levinthal & March, 1993; Nelson & Winter, 1982) or enabling change by leveraging prior related competence or experience (Cohen & Levinthal, 1990; Helfat, 1997; King & Tucci, 2002). The question for research on managerial cognition, therefore, is how cognitive effects relate to those of organizational incentives and capabilities.

This study focuses on the cognition of CEOs specifically and examines the influence of CEO attention relative to the effects of these important organizational factors in shaping firm response to technical change. The empirical context is the fiber-optic revolution in the communications technology industry. Optics were potentially competence-destroying (Tushman & Anderson, 1986) for firms supplying the existing copper-based technologies, and investments in this new technical arena were both costly and risky. Therefore, CEOs faced a great deal of uncertainty about whether or not to invest. The panel data set covers 71 firms, containing observations of both

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incumbents and entrants over two decades, from the early introduction of fiber optics through the internet “bubble” and crash (1982–2001). Using such a longitudinal approach focuses analysis on how changes in the attention CEOs placed on optics led to subsequent changes in firms’ technical directions.

The analysis proceeds through the estimation and interpretation of a diffusion equation in which the dependent variable is a measure of the extent to which a firm invested in optics and the key explanatory variable is a measure of the attention that a firm’s CEO paid to optics in a prior year. I compare this with the effects of organizational incentives and prior related capabilities. A finding that a CEO’s attention is both distinct from but interrelated with organizational explanations of firm response to technical change would suggest that managerial interpretive processes are an important arbiter between the environment and the strategic actions of the firm. By considering how cognition, capabilities, and incentives interact, the analysis in this article develops a more contingent view of the conditions under which CEO cognition might matter.

THEORETICAL BACKGROUND AND HYPOTHESES

A cognitive explanation of firm response to technical change rests on the notion that firms’ actions are shaped by how managers make sense of their environments (Thomas, Clark, & Gioia, 1993). Managerial cognition thus connects a firm’s actions to a changing environment by influencing what is noticed, how this information is interpreted, and why certain choices are made (Daft & Weick, 1984; Ocasio, 1997). In periods of relative stability, the interpretations existing among firms are likely to converge (Porac, Thomas, & Baden-Fuller, 1989; Spender, 1989). Within any one firm, they are also likely to become entrenched in a set of routines that link cognition, capabilities, and incentives together. However, in periods of technical change, when uncertainty is high, these heuristics break down. As a result, different firms’ interpretations are likely to diverge widely, and tight alignments between cognition, capabilities, and incentives within firms likely break apart (Bogner & Barr, 2000; Reger & Palmer, 1996). The firms that successfully respond seek a new alignment among these factors.

A simple model of these effects would be one in which firm adaptation to change is most dramatic when cognition, capabilities, and incentives are all aligned toward a new opportunity and least effective (or nonexistent) when these factors are locked into old ways of doing business. A more complex model

would also take into account the relationships among these managerial and organizational factors. Are they simply individually important but separate drivers of outcomes? Or does the presence of organizational capabilities or incentives moderate the effect of managerial cognition? And what about situations in which organizational capabilities or incentives are lacking? Can managerial cognition directed toward a new technology substitute for gaps in the necessary organizational enablers of change?

The underlying premise of this study is that cognition, capabilities, and incentives are not just alternative explanations of organizational outcomes, nor are they merely unrelated forces shaping the adoption of a new technology. Rather, these managerial and organizational factors are meaningfully linked. Yet these links have yet to be systematically explored in the literature (McGee, 2005). This article contributes to an understanding of the relationships among cognition, capabilities, and incentives through an examination of their effects when considered independently and, more importantly, in interaction.

The Role of Managerial Cognition

Previous research has established the importance of senior managers in shaping strategic action during eras of upheaval (Eisenhardt & Bourgeois, 1988; Hambrick & Mason, 1984; Tushman & Rosenkopf, 1996; Virany et al., 1992). Top team demographic factors such as tenure and background have been associated with the ability to navigate radical technical change (Lant, Milliken, & Batra, 1992; Tushman & Rosenkopf, 1996; Virany et al., 1992). Researchers have used CEO tenure and functional background to capture constructs such as receptivity to change and willingness to take risks (Datta, Rajagopalan, & Zhang, 2003; Wiersema & Bantel, 1992). CEOs with technical backgrounds may be more likely to understand the value of new technical arenas (Barker & Mueller, 2002). CEOs with outside experience have also been found to be receptive to new ideas, and CEOs with greater tenure in a firm have been found to be more likely to resist new technologies because they get embedded in the firm’s traditional way of thinking about its business (Finkelstein & Hambrick, 1990). On the other hand, greater tenure in the CEO job could give CEOs both the freedom to be more risk-taking (because they might no longer need to prove themselves in the short term) and the power to drive their firms in new directions.

Many of the studies cited above make implicit reference to cognitive processes (e.g., receptivity), primarily using demographic measures as proxies (Hambrick & Mason, 1984; Norburn & Birley, 1988;

Wiersema & Bantel, 1992). Demographic characteristics are important indicators of sources of cognition—they can reasonably be seen to represent managers' knowledge accumulations from prior experiences—but they cannot capture the context-specific interpretation of a particular technical change. Indeed, research has suggested that demographic traits are weak approximations of cognition, and therefore more direct measures should be considered (Barr et al., 1992; Markoczy, 1997).

Some case studies have more directly addressed the role of top managers' cognition in shaping firm strategic action in the face of industry change. Barr, Stimpert, and Huff (1992) studied two railroad companies and found that managers' evolving views, along with their ability to connect these views to action, explained the firms' differing responses to changes in the industry over time. Barr's (1998) analysis of several pharmaceutical firms' "cognitive maps" demonstrated that the degree of change in the maps affected the timing of response to change in the industry. Tripsas and Gavetti (2000) showed that Polaroid's paradoxical response to the transition from analog to digital imaging technologies (aggressive development of digital technologies but failure to alter its business model) was a consequence of a mismatch between executives' belief structures and the digital world.

Research using a cognitive approach to understanding strategic change has often exhibited the difficulty of distinguishing between cognitions and economic outcomes in defining constructs that can be readily compared across studies (Rajagopalan & Spreitzer, 1997). An increasing number of quantitative studies on managerial cognition (e.g., Baum & Lant, 2003; Porac, Thomas, Wilson, Paton, & Kanfer, 1995; Thomas et al., 1993) have contributed to a menu of ways to measure cognition more systematically. These studies have, however, generally focused on stable settings or used cross-sectional data, which makes it more difficult to separate cognition from other explanations and to examine its effect on subsequent action.

Some recent studies have addressed these gaps by using panel data sets that exploit both cross-sectional and longitudinal variation to examine the effect of cognition in response to industry change. Kaplan, Murray, and Henderson's (2003) analysis of 15 large, incumbent pharmaceutical firms' responses to the emergence of biotechnology and Cho and Hambrick's (2006) study of airline deregulation demonstrate that a CEO's attention to environmental changes can shape organizational investment in new arenas. What remains, then, is to explore managerial effects on investments in new technologies

that go above and beyond those of organizational incentives and capabilities.

Hypothesis 1. Other things being equal, increases in CEO attention to a new technology are associated with subsequent increases in a firm's investment in that technical domain.

Organization-Level Explanations of Firm Response to Technical Change

The view that CEO cognition matters for firm response to technical change must be examined alongside organization-level explanations of action already well established in the literature. Although such explanations have, in the main, been explored independently from that of cognition, the argument made in this article is that CEO- and organization-level factors must be considered jointly if one is to understand strategic action in the face of change.

Broadly, these other explanations are either primarily economic or primarily behavioral and capabilities-based. Much of this research has been intended to explain the outcomes of battles between incumbents and entrants. Studies have shown how, in a vast array of industries, new entrants displaced incumbents as market leaders when a new technology invaded a market (cf. Christensen & Rosenbloom, 1995; Cooper & Schendel, 1976; Henderson & Clark, 1990; Utterback, 1994), though other research has shown that incumbents can survive if they invest in the right technologies or have "complementary assets" (Christensen, Suarez, & Utterback, 1998; Tripsas, 1997).

Scholars have made economic arguments that differences in organizational incentives can explain these outcomes. Under some conditions, incumbents will invest less than entrants because of a fear of cannibalizing their existing "rents" (Cockburn & Henderson, 1994). Alternatively, dependence on a core set of customers may create incentives for incumbents to lead rather than lag (Mitchell, 1989). Resource-dependence theory suggests that established firms get trapped by their existing customer bases and therefore can only respond to new technologies to the extent they are useful to those market segments (Christensen & Bower, 1996). When a firm depends on a set of customers for its profits, the organizational incentives to invest in new technologies that do not appeal to those customers are relatively low. Conversely, if the existing customers are interested in the new technology, an incumbent has higher incentives to act. In addition, investments in a new technology can protect a core market by creating user "switching costs" (Lieberman & Montgomery, 1988) or preempting competition (MacMillan, 1983).

Hypothesis 2. Other things being equal, firms with higher incentives to adopt a new technology (those whose existing customers desire it) engage in higher levels of investment in that technical domain.

A more behavioral point of view (Nelson & Winter, 1982) suggests that routines and capabilities govern whether incumbents or entrants survive and perform (Sorensen & Stuart, 2000). Incumbents may experience a substantial liability of aging because rigidification of routines leaves them unable to respond to a new context (Barnett, 1990) in which their core competencies may not be useful (Leonard-Barton, 1992; Levinthal & March, 1993). Alternatively, this honing of routines and capabilities may allow firms to function well, thus leading entrants to experience a liability of newness (Hannan & Freeman, 1984; Stinchcombe, 1965). However, theories of absorptive capacity suggest that it is not incumbency per se, but rather the degree to which prior competence is related to achieving success in a new technology, that conditions investment in the new arena (Cattani, 2005; Cohen & Levinthal, 1990; Helfat, 1997).

Such prior related competence can be thought of as accumulation of experience in an affiliated domain. Accumulations of experience associated with the new technical arena should help firms adapt over time (Mitchell & Singh, 1992; Rosenkopf & Nerkar, 2001). Technical experience has been shown to be associated with research productivity (Henderson & Cockburn, 1994) as well as with the likelihood and success of new-product or new-market entry (Katila & Ahuja, 2002; King & Tucci, 2002; Martin, Swaminathan, & Mitchell, 1998; Nerkar & Roberts, 2004). This research suggests that accumulations of related experience should be expected to enable response to technical change.

Hypothesis 3. Other things being equal, firms with greater accumulations of prior related capabilities (experience) in a new technology engage in higher levels of investment in that technical domain.

Although none of these incentives- and capabilities-based explanations explicitly models cognition, one could easily reinterpret them using a cognitive lens. For example, is it dependence on the rents from key customers per se that causes firms to miss new technologies? Or perhaps this failure stems from the fact that CEO attention is focused on one arena to the exclusion of others? Do existing capabilities lock firms in, or is it the inability to change how managers see the world? It is therefore useful to explore cognition in relation to these organizational factors. No existing quantitative stud-

ies explicitly measure and interact cognition, incentives, and capabilities, yet there is reason to believe that such interactions are important.

Recent qualitative and theoretical work has begun to explore the separate but intertwined effects of cognition and economic or incentives-based factors in shaping organizational evolution. On the one hand, cognition and incentives have been portrayed as substitutes for each other. As Burgelman (1994) demonstrated in the case of Intel, middle managers pursued firm-level interests to move to a new technical arena despite CEO inattention to the issue. In a scenario such as this, firms invest in new technologies because of firm-level incentives to do so, even in the absence of related managerial cognition. The implication is that as incentives increase, the effect of CEO attention is attenuated.

On the other hand, cognition and incentives have also been shown to be complementary. In subsequent analyses of Intel strategy making over the years, Burgelman (2002) noted that top managers' attention to a new technology can amplify the effect of organizational incentives on change. This view is consistent with the idea that the organizational routines associated with making investments in a new technology, for example, require a truce between the routines' cognitive and incentive components (Cohendet & Llerena, 2003; Dosi, Levinthal, & Marengo, 2003). This architecture is the result of a tight alignment of cognition with an understanding about what kind of effort gets rewarded (Kaplan & Henderson, 2005). If aligned toward the new opportunity, this truce can enable adaptation. But inertia can result if this organizational architecture is locked into old ways of doing business (Henderson & Clark, 1990). Economic forces may reinforce "attention traps" (Johnson & Hoopes, 2003: 1058) that cause managers to miss out on the big picture of change.

Thus, although the view that cognition and incentives are interwoven is emerging, research to date provides little definitive indication about the direction of the relationship. As hypothesized above, the main effects of both cognition and incentives on subsequent adoption of a new technology should be positive, but the nature of their conditional effects is less clear. That is to say, the question remains, Will the effect of CEO attention to a new technology be diminished or amplified by increases in incentives to adopt? Thus, I propose two competing hypotheses about the interaction of cognition and incentives:

Hypothesis 4a. The effect of CEO attention to a new technology on a firm's investment in that

technical domain is a decreasing function of the firm's incentives to invest in that domain.

Hypothesis 4b. The effect of CEO attention to a new technology on a firm's investment in that technical domain is an increasing function of the firm's incentives to invest in that domain.

A similar argument can be made about the relationship between cognition and capabilities. Broadly, there is evidence that top managers moderate the effect of experience (Adner & Helfat, 2003; Eisenhardt & Martin, 2000; King & Tucci, 2002), though different scholars have produced diverging results on the direction of this conditional effect.

Managerial efforts to change strategic direction have been shown to substitute for the organizational capabilities necessary for adaptation. Rosenbloom (2000) suggested in the case of NCR that top managers compensated for a lack of organizational capabilities by breaking with the past and redirecting the firm. Managerial cognition can therefore be seen as guiding the subsequent adoption of a technology and development of related capabilities (Gavetti & Levinthal, 2000; Gavetti, Levinthal, & Rivkin, 2005). Ex post, cognition and capabilities appear to have been aligned to the new opportunity, but ex ante, cognition preceded the development of capabilities. As capabilities increase over time, the effect of cognition may therefore be attenuated.

Yet it has also been argued that cognition and capabilities are complementary and operate simultaneously on strategic action. If they are appropriately aligned with an opportunity, they can work together to accelerate change (King & Tucci, 2002). Without such links, inertia may result. The misalignment of top managers' cognition and organizational capabilities can impede a firm's ability to invest in a new domain (Tripsas & Gavetti, 2000).

This evidence supports the idea that cognition and capabilities are intertwined, but again, the direction of the conditional effect is not clear. Where organizational rigidities impeding change exist, cognition can either overcome or amplify their effects, depending on the circumstances (Gilbert, 2005). Whether the presence of related organizational capabilities dampens or amplifies the effect of CEO attention on adoption of a new technology remains an open question.

Hypothesis 5a. The effect of CEO attention to a new technology on a firm's investment in that technical domain is a decreasing function of the firm's capabilities (experience) in that domain.

Hypothesis 5b. The effect of CEO attention to a new technology on a firm's investment in that technical domain is an increasing function of the firm's capabilities (experience) in that domain.

Testing this series of hypotheses should help unpack the separate and potentially related effects of cognition, capabilities and incentives on firm response to technical change.

METHODS

Research Setting

This study examines communication technology suppliers' response to optical technologies during the fiber-optic revolution. The firms in this industry produce the components and systems that make up communication networks. Although its scientific roots are in discoveries made throughout the 20th century (and before), optics began to influence the direction of the communications industry in the 1980s.

The communications technology industry was an appropriate setting in which to examine the dynamics of firm response to technical change. The emergence of optical technologies involved a dramatic improvement of technical performance, triggered a proliferation of new firms, changed the fate of incumbents, and altered the economics of the industry (Alferness, Kogelnik, & Wood, 2000; Hecht, 1999; Solymar, 1999). Specifically, optics was a radical technology that was competence-destroying (Tushman & Anderson, 1986) for incumbent communications firms that had developed extensive capabilities in analog and digital transmission over copper wires. Optics, however, was not necessarily "disruptive" in the narrow sense (a disruptive technology is one for which a firm's current customer base has little immediate use [Christensen & Bower, 1996]). Optics was important and useful for telecommunications carriers (e.g., Verizon, MCI, British Telecom, Bell Canada) whose challenge it was to improve the quality of connections for voice communications. Therefore, any communications technology firm already serving "telecoms" (telecommunication) carriers had strong incentives to invest in optics.

Ex post, investment in optical technologies has generally been seen as beneficial for communications technology suppliers (especially those serving telecoms carriers), yet ex ante there was a great deal of uncertainty about whether these technologies should be adopted, especially given the very high expense and risk associated with such investment and their competence-destroying effect. Communication technology firms' responses in the form

of optical patenting varied widely. The purpose of this study was to explore the extent to which CEO cognition could explain this heterogeneity.

The fiber-optic revolution took place over two decades, generating several waves of uncertainties as new entrants emerged, market structures changed, and enabling technologies were introduced. The first commercial communications applications of simple optical technologies came in the mid to late 1980s. Yet the expansion of early systems was limited by the cost of amplifying signals, raising doubt about the viability of optics. However, the breakup of AT&T in 1985 triggered a wave of new customers in the form of the "Baby Bells," which created fertile ground for firms selling optical technologies. In addition, the years 1991 and 1992 saw the invention of new enabling technologies in the form of erbium-doped fiber amplifiers (EDFAs, an all-optical amplification system) and wavelength division multiplexing (WDM), which made long-distance systems less expensive than earlier optical systems and increased the capacity of each fiber by allowing multiple wavelengths of light to be sent down the same strand.

The introduction of multiplexing technology combined with the emergence of the internet led to high growth in communications in the late 1990s. The Telecommunications Act of 1996 deregulated the industry in the United States and paved the way for the entry of many CLECs (competitive local exchange carriers). These carriers were not burdened with legacy equipment and were therefore even more likely to buy optical technologies. Many new suppliers entered to serve these needs. These factors combined to create the communications bubble of the late 1990s. But the forces of increasing bandwidth collided with rampant programs to lay fiber, leading to a situation of dramatic oversupply. These market dynamics burst the bubble of overinvestment and triggered the subsequent market crash in 2000–01. Here, I examine communication technology firms' investment decisions over these 20 years of turbulence.

Sample

The analysis covers 71 communications technology suppliers during the period 1982–2001. They represent a stratified random sample drawn from 617 firms traded on U.S. stock exchanges, about 10 percent of which are non-U.S. firms. I established the original population by selecting the 1,752 firms listed in Compustat in primary and

secondary NAICS codes¹ covering communications firms over the 20-year period of the study. As the NAICS codes are quite broad, a shorter list of the firms predominantly in the communications industry was established. Ideally, one would use the percentage of sales in the communications industry to filter out unrelated firms. However, this information is only available sporadically and in inconsistent formats. Where reliable data could be located, I aimed to select firms with more than 25 percent of their sales in communications. To supplement this information, three coders used descriptions of each firm from the Compustat database, Securities and Exchange Commission (SEC) 10K filings, and *Moody's Industrial Manual* to determine whether a firm had a significant communications technology business.² The combination of these methods produced a population of 617 firms.

Because the management of technology literature has emphasized different outcomes for incumbents and entrants and for large and small firms, I used a sampling method that would assure that the sample represented firms of all types. Thus, from the population of 617, I selected 71 through stratified random sampling based on timing of entry (incumbents and entrants from different eras) and size of firm. The 617 firms were divided into four entry cohorts—incumbents (entry 1985 or before), early entrants (1986–91), mid entrants (1992–96), and bubble entrants (1997–2001)—that correspond to the critical industry turning points discussed above. They were also categorized into three size cohorts by level of annual sales: large (greater than \$1 billion in average inflation-adjusted sales over the time period in the data set), medium (\$100 million to \$1 billion), and small (less than \$100 million).

Where possible, 5 firms were randomly selected from each of the 12 strata of firms (four age cohorts times three size cohorts). I also oversampled the population of large incumbents, initially selecting all 20 firms in order to have a larger sample for incumbent-entrant comparisons. Dropping Perkin Elmer from the analysis because of problems of data matching led to a sample of 19 incumbents. Ten firms exited the data set by being acquired or

¹ The NAICS (North American Industrial Classification System, previously known as the Standard Industrial Classification [SIC]) codes covering the communications industry include 334119, 334210, 334220, 334413, 334418, 334415, 335921, and 541512.

² Interrater reliability was .75. Differences were resolved through discussion and a further review of company materials.

going bankrupt. Analyses conducted without these firms in the sample were consistent with overall findings, and thus I believe these exits did not introduce bias into the results. The Appendix includes a list of the companies in the sample. Of special note, the sample includes Bell Labs (first under AT&T and then later as part of Lucent), one of the most prolific patenting organizations in telecommunications. The results reported in this article are robust to the exclusion of AT&T and Lucent. The final sample of 71 firms included 822 potential firm-year observations (though the lag structure used in regressions and data availability issues reduced the observations to 528 for reported results).

Analytical Approach

The empirical strategy was aimed at teasing apart the effects of cognition, capabilities, and incentives in shaping communication technology firms' responses to optics. To do this, I estimated and interpreted a simple diffusion equation:

$$Y_{j,t} = \alpha_j + \beta_1 X_{j,t-n} + \beta_2 C_{j,t-n} + \beta_3 I_{j,t-n} + (\beta_4 X_{j,t-n} \times C_{j,t-n} + \beta_5 X_{j,t-n} \times I_{j,t-n}) + \beta_6 Z_{j,t-n} + u_j + \varepsilon_{jt},$$

where $Y_{j,t}$, the dependent variable, is a measure of investment in a new technology (the extent to which firm j responded to optical technologies at time t). The key explanatory variable $X_{j,t-n}$ is a measure of the attention that the CEO of a particular firm paid to optical technologies, $C_{j,t-n}$ is the measure of related capabilities, $I_{j,t-n}$ is a measure of firm-level incentives to invest in optics, and $Z_{j,t-n}$ is a vector of control variables, each of these in a prior year. To understand whether cognition was dependent on either capabilities or incentives, I also introduced interactions. In addition to the standard error term, I also included random effects to correct for a lack of independence across firm observations (u_j).

I expected that changes in a CEO's attention to optics would be systematically associated with subsequent changes in investment patterns in the form of optical patents, even when accounting for organizational factors. The effects were tested over various time lags (intervals of from one to four years between mentions of optics in firms' "letters to stockholders" and patenting). To alleviate concerns that it might be patenting that led to CEO attention or that there was another cause, such as an exogenous shift in the market driving both patenting and cognition, I also tested for reverse causality.

Dependent Variable: Measure of Investment in Optics

The study assesses the effect of CEO attention to optics on subsequent strategic action in the form of investment in optical technologies. The preferred way to measure technical investment would be to use R&D expenditures for optical technologies, but firms do not make these program-specific data available. Instead, in this analysis I used patents as a proxy. Patents are often used in strategic analysis to measure the degree of innovative outcomes (Griliches, 1990), particularly when weighted by citations to indicate impact (Jaffe & Trajtenberg, 2002). However, it is also credible to use patents as a measure of innovative effort or investment because they can be seen to reflect work directed toward the creation of a new product (Schmookler, 1972). Patenting in specific technical arenas has been shown to be correlated with program-level R&D (Henderson & Cockburn, 1994). I conducted interviews with engineers from a number of communications technology firms as a preliminary to this study; qualitative evidence from these interviews indicated that, in the context of this industry, the choice to patent can reasonably be thought of as an indicator of the level of investment in that arena. Patent ideas generally come out of project work, so they are more likely to emerge from officially sanctioned and funded activities. It thus appears that patents should be a reasonably good proxy for strategic choices about investment allocations in this engineering-intensive industry.³

Because the dependent variable in this analysis had to represent investment associated with optics, I measured optical patents as a percentage of total firm patents in order to capture the emphasis placed on optics. This percentage was calculated as a count of all optical patents (based on primary patent class) in the application year of each patent, normalized by a count of a firm's total patents in all classes.⁴ Although the mean value of this measure

³ This situation stands in contrast to that in more science-oriented industries, such as pharmaceuticals, where scientific publications might also be considered a proxy for investments in a new technology (Cockburn & Henderson, 1998). Scientific publishing in the communications technology industry was extremely limited and typically not encouraged for fear of releasing proprietary information that would allow competitors to copy or reverse-engineer a technology.

⁴ There is always a delay between the application and the granting of a patent by the USPTO (the average in these classes is just over two years). Therefore, the application date is the better representative of the timing of a

of the dependent variable is 5 percent, there is a good deal of variation, as the range is from 0–100 percent and the standard deviation is 13 percent. The patent data come from the U.S. Patent and Trademark Office (USPTO) website. The three-digit USPTO classes associated with optics in communications are 356 (optics: measuring and testing), 359 (optical systems and elements), 372 (coherent light generators), and 385 (optical waveguides).

Main Independent Variable: Cognitive Measure of Attention to Optical Technologies

Constructing a direct measure of managerial cognition has been problematic for researchers, particularly for those desiring to conduct statistical analyses (Lant & Shapira, 2001). One must construct a variable from data that are consistent across firms and over time and amenable to quantitative measurement. Content analysis of management texts has been acknowledged as a meaningful way of capturing managerial cognition and is particularly useful in longitudinal research designs (Duriau, Reger, & Pfarrer, 2007; Walsh, 1995). Following recent management scholarship (Cho & Hambrick, 2006; Kaplan et al., 2003), I used a count of optics-related words appearing in the letters to shareholders of each company to capture a CEO's attention to optics. A letter to shareholders bearing the signature of a firm's CEO is a regular part of its annual report. Below, I address two concerns one might have about this approach: the appropriateness of letters to shareholders as the source material and the usefulness of word counts per se.

With regard to the use of letters to shareholders, substantial evidence suggests that they are reasonable sources of data for measuring CEO attention (cf. Barr, 1998; Barr et al., 1992; Osborne, Stubbart, & Ramaprasad, 2001). Other kinds of statements by CEOs, such as those obtained through interviews or surveys, might initially appear to be attractive sources, but they are impractical for larger samples of firms over long periods. Such sources might be subject to the risk of retrospective bias; managers

would likely adapt their memories of their views in prior years to subsequent outcomes. Given the long periods of time involved in this analysis and the dramatic shifts that occurred in the focal industry, it seemed unlikely that such bias could be adequately mitigated. As a result, it was imperative to locate consistent sources of information on CEOs' contemporary, prospective views across firms and over time. The letters to shareholders were the only source of information about CEO attention that met this stringent test.

Some have argued that letters to shareholders may be outcomes of public relations work or symbolic management activities and therefore decoupled from leaders' real actions (Fiss & Zajac, 2006); however, both theoretical and empirical rationales suggest one can construe these documents to be important evidence of CEO attention.

First, although it is the case that the bulk of each annual report is written by the communications department in a firm, evidence substantiates that the letter to shareholders itself is written or closely reviewed and edited by the CEO (Kohut & Segars, 1992). The letter goes out under the CEO's signature, and therefore, for fiduciary reasons, it is unlikely that the CEO would suppress discussions of important issues in it, even if they did not reflect favorably on the company. Indeed, the SEC Act of 1934 constrains executives to avoid "any untrue statement of a material fact or to omit to state a material fact" (from Rule 10b-5), including in the letter to shareholders (Abrahamson & Park, 1994).

Second, Fiol's (1995) study comparing internal views (as represented by strategic planning documents) and external views (as represented by annual reports) showed that although judgments about the valence of an issue (for instance, as a threat or as opportunity) were not significantly related in the two kinds of documents, the basic thematic emphases were the same. Taken together, this evidence suggests that letters to shareholders can be construed to be a reasonable proxy for CEO attention to what is important for a company's performance and future prospects, but that their use in this context should focus only on broad themes rather than on valence.

These themes can be manifested in the specific words used in a text (Cho & Hambrick, 2006). Research has shown that words can be taken to be representations of cognitive categories, and more frequently used words can be taken to be indicators of what categories actors find important (Duriau et al., 2007). Drawing on this understanding, I chose to use counts of the optics-related words in a letter to capture CEO attention to optics. These counts were normalized by the total number of words in

strategic action taken by a firm. However, the application dates are only available for patents that are ultimately granted, leading to potential undercounting of actual patent applications. This limitation only makes this analysis a conservative test of the hypotheses. The patent search was done in October 2003; therefore, there may be truncation of patent counts in the latter years of the data set. The effects of undercounting and truncation are minimized when the dependent variable is normalized by total patents granted per firm.

the letter to produce a measure of the emphasis on optics relative to other themes. This measure of frequency is a proxy for CEO attention (Abrahamson & Park, 1994; Wade et al., 1997). Such a thematic approach to textual analysis was also the most appropriate given the need to generate quantitative measures over a large number of texts (Roberts, 1997; Walsh, 1995). As Osborne, Stubbart, and Ramaprasad noted, "Variations in theme content or frequency [in a letter to shareholders] may reflect changes in leadership perspectives and highlight observable shifts in subsequent performance. . . . [They] represent outward manifestations of company intentions" (2001: 238, 240). I therefore concluded that word counts were a reasonable proxy for attention to a particular theme. The strength of the use of word counts from the letters to shareholders was that they were prospective, consistently available, and replicable (Bettman & Weitz, 1983).

A coding list of optics-related words (i.e., words, phrases, abbreviations, prefixes, and their variants) was developed from the ground up. Included were "EDFA" (erbium-doped fiber amplifier), "fiber," "fiber-optic," "laser," "light," "wave division multiplexing," "optic," "opto-," "photonic," "SONET," "SDH," "lightwave," and "waveguide."⁵ I selected these terms after reviewing a concordance of the entire list of more than 26,000 separate words that appeared in the letters to shareholders in this sample. Coding was done in such a way as to identify the context-specific use of words. For example, "light" was counted only when referring to optics and not when used in a phrase such as "in light of." For the 822 firm-year observations, I was able to locate 719 letters. Another 103 letters could not be located and were likely not published. These firm-year observations are not included in the analysis. I lagged these counts to account for the fact that responses to changes in CEO attention are not instantaneous: some time is required for an organization to take action and apply for patents.

Related Factors Shaping Firm Response to Technical Change

As mentioned in the theory section, organizational incentives and capabilities have predominated in the management of technology literature as explanations for heterogeneous firm responses to technical change. I therefore included them in this

analysis in order to consider their relative and interrelated effects with cognition.

Organizational incentives. Dependence on an existing customer base can create powerful incentives to invest in new technologies if such technologies are demanded by the customers or if action is required to preempt competition or create user switching costs (Christensen & Bower, 1996; Lieberman & Montgomery, 1988; MacMillan, 1983; Mitchell, 1989). An examination of different types of customers can help elucidate these types of organizational incentives. Firms selling communications technologies have two major classes of customers. Many suppliers focus on serving carriers (such as Verizon, MCI, or British Telecom) that buy voice-related products and demand the high technical standards of "carrier grade" products. Other suppliers focus on enterprises (corporations and other large organizations) buying data-related products that only have to enable lower-tolerance "best effort" connections. Users of data in enterprises do not observe small delays or errors in reassembling data packets; the same cannot be said for even very small flaws in voice connections provided by carriers. It was likely, then, that communications technology firms primarily serving carriers, with their higher quality requirements, would be more likely to be interested in pursuing optical technologies that could transmit more information at higher speeds. As such, serving carriers would provide higher incentives to communications technology suppliers to invest in optics.

Because firms do not provide information on sales to carriers in any consistent or reliable way, I developed careful estimates of these data. I gathered communication sales information for each company based on segment data from Compustat where available. This was supplemented by reports of communications sales from company press releases, media reports, and analysts. To understand the portion of these sales that were specifically made to carriers, I adjusted this number using estimates of emphasis on carriers based on coding by three industry experts.⁶ This procedure allowed me to estimate the amounts of total sales coming from carriers only. The higher this number, the greater a

⁵ SONET (synchronous optical network) and SDH (synchronous digital hierarchy) are both standards for optical communications transport.

⁶ Coder 1 was in the advanced technologies and business development group of a major equipment company. Coder 2 was an industry analyst. Coder 3 was a materials science professor active in the communications industry. Average interrater reliability was .72. I resolved any differences in coding in a thorough review of company information, including SEC filings and analyst reports.

firm's incentive to invest in optics. This value was logged and entered with a three-year lag.

Organizational capabilities. Resource-based views suggest that prior related competence should condition future investment in new technologies (Cockburn, Henderson, & Stern, 2000; Cohen & Levinthal, 1990; Helfat, 1997). Here I focus on capabilities in the optical product-market, a construct that is primarily a function of time and can be meaningfully measured by the cumulative number of years of experience in a particular arena (King & Tucci, 2002; Levin, 2000). The variable used was a measure of the number of years a firm had had an optical product in the market. The data for the timing of launch of the first optical product were developed from company new-product announcements and the *CorpTech Directory*, which lists key product lines, using survey responses by participating firms.⁷ This variable was also entered with a three-year lag.

Controls

I included an additional set of variables that controlled for other firm-related factors associated with investment in a new technology. Each of these was entered with a three-year lag, except for two non-time-varying firm characteristics (focus on cellular technologies and timing of firm entry).

Scale. There may be economies of scale in the adoption of new innovations whereby additional investments in new technologies lead to efficiencies (Cockburn et al., 2000; Zucker & Darby, 1997). The analysis included the log of total firm sales (as obtained from Compustat) to control for potential economies of scale. Scale could alternatively be measured by R&D spending to capture the size of a firm's technical effort. Indeed, sales and R&D spending were highly correlated (.92) and independently were significantly associated with optical patenting. The results were robust to the inclusion of the log of R&D spending in lieu of sales.

Financial resources. Operating income as a percentage of sales (as obtained from Compustat) measured the extent of financial well-being. Higher returns were a measure of the availability of firmwide resources. It can be assumed that such financial

slack (Bourgeois, 1981; Thompson, 1967) in an organization could be used for investment in a new technological field.

CEO demographic characteristics. Because CEO demographic characteristics have often been used as proxies for cognition, I chose to include several measures of CEO background as controls to help elucidate the context-specific interpretation (as represented by CEO attention) relative to more general demographic influences on receptivity to change. These measures of receptivity included functional background (technical or other), tenure in the current firm, tenure in the current CEO position, and a dummy for CEO experience outside of the focal firm (Norburn & Birley, 1988; Virany & Tushman, 1986; Wiersema & Bantel, 1993). These data came from a variety of sources, including company press releases, *Who's Who*, SEC filings, company websites, *Moody's Industrial Directories*, *CorpTech Directories*, and phone and e-mail requests to the companies. Owing to problems of collinearity between CEO firm tenure and CEO outside experience ($-.69$), I omitted the latter from the regression equations (but the results are robust to replacing tenure with experience).

Focus on cellular technologies. Firms in the communications technology field during the period covered in this study could have reasonably chosen to focus on cellular (wireless) technologies rather than fiber optics. Because these two technical arenas are quite different, those firms choosing cellular technologies would be much less likely to emphasize optical technology investments. I therefore included a control for a firm's focus on cellular technologies, a dummy variable coded 1 if a firm was in the cellular arena.

Timing of entry. Because research has suggested that incumbents and entrants will respond differently to technical change (for a summary, see Utterback [1994]), I included two controls for the timing of entry, one for early entrants (firms that entered between 1986 and 1991) and one for later entrants (firms that entered after 1991). Note that the middle entrants and bubble entrants in the sample are aggregated here as the number of firm-year observations for the latest entrants was small.

Time trends. Because many confounding events occurred during the evolution of optics, it was important to control for the effect of time. Several time-varying factors could have shaped firms' responses to change. The first was demand. Some research in the management of technology field has suggested that demand is the critical driver of innovative output (Schmookler, 1972). Here, I measured demand using data from the Value Line Investment Survey on U.S. communication carriers'

⁷ Product line codes were PHO-AO (acousto-optic equipment), PHO-FO (fiber optics, related equipment), PHO-LA (lasers, laser-related equipment), PHO-OE (opto-electronic equipment), and PHO-OP (optics, related equipment). CorpTech data are only available from 1986, so timing of entry for incumbent firms was based on press releases and company data.

capital expenditures. A measure of demand would ideally focus on customers' spending on optical technologies, but these data were not available for all of the years in the data set (most analysts initiated coverage of the optical segment in the mid 1990s), so a broader proxy was used. Second, institutional arguments would suggest that strategic response might be driven by pressure toward isomorphism, either from competitors or from market trends overall (DiMaggio & Powell, 1983). To control for the possibility of competitive isomorphism, I included competitive actions as measured by a count of all optical patents for each year at each application date (from the USPTO). Counts of *Wall Street Journal* articles mentioning optics measured market trends. The *Wall Street Journal* is often cited as the newspaper of record in business and therefore such an article count is a reasonable proxy for trends in the market. Further, Sine, Heman, and Tolbert (2005) argued that media mentions represent the legitimacy of a new market segment. Because the *Wall Street Journal* is widely read not only by managers but also by investors, coverage of optics in this newspaper would likely create isomorphic pressures for a broad set of industry actors.

These three measures of time (data on demand, patents, and media mentions) were reasonably highly correlated (correlations were all greater than .55). Therefore, because the count of articles in the *Wall Street Journal* was the most analogous to the measure used for the independent variable of interest (cognition as measured by word counts in letters to shareholders), I used it in the regression analyses to capture the effect of the time-dependent factors. The results are robust to the use of the alternative measures, as well as to the use of year fixed effects.

Lagged dependent variable. I include a lagged dependent variable in all models. My qualitative knowledge of communications technology firms indicated that research programs have a certain amount of momentum and do not fluctuate wildly from year to year. This view is consistent with research on patenting patterns in other contexts such as pharmaceuticals (Cockburn et al., 2000).

Random effects. To correct for unobserved heterogeneous firm characteristics that might shape response to optical technologies, I used a random-effects specification. This method was used instead of a fixed-effects specification so that non-time-varying variables, such as customer type, timing of entry, and focus on cellular technologies, could be

included (Greene, 2003).⁸ The results are robust to the use of firm fixed effects (with the non-time-varying variables omitted). By including random effects and the lagged dependent variable in the analysis, I was able to look at how changes in CEO attention to optics affected subsequent changes in patenting.

Estimation

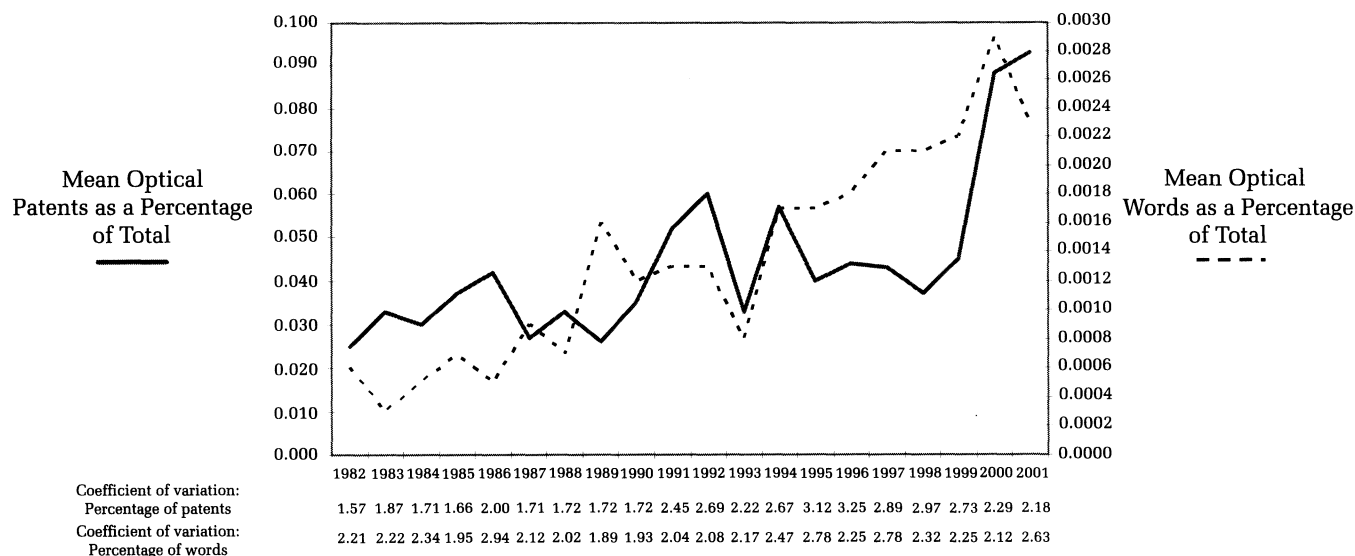
One important estimation issue when using optical patenting as a percentage of total patents is the presence of a large number of zero outcomes. A zero may represent a simple lack of optical patents in a single year or may be the result of a lack of optical patenting (or any kind of patenting) altogether. In this sample, 88 percent of firms had at least one year with no optical patents; 56 percent of firms had no optical patents in any year; and 8 percent had no patents of any kind in any year in the period studied. However, all firms in the sample listed R&D expenses in nearly all years. Zeros could represent a lack of adequate investment or a lack of success. To the extent that the zeros represent unsuccessful attempts or deliberate choices not to patent, a censored Tobit regression model could compensate (Greene, 2003; McDonald & Moffitt, 1980). Use of censored Tobit was appropriate as the independent variables could predict both the probability and the degree of patenting. Here, the latent dependent variable represented an individual firm's unobserved propensity to patent in the optical arena. Separately, I analyzed absolute optical patent counts in a negative binomial count model. The results are robust to this specification (details are available from the author).

RESULTS

In this section, I report on a series of analyses designed to explain optical patenting among communication technology suppliers, and in particular to examine the effect of CEO attention on shaping these patterns over time. On average, an emphasis on optical patenting as well as mentions of optical technologies in the shareholder letters increased over time for the firms in the data set, as can be seen in Figure 1. The initial evidence graphed in this

⁸ Note that there are some concerns about using random-effects models when it is possible that individual effects are correlated with other regressors, such as the lagged dependent variable (Greene, 2003). The results of models without the inclusion of random effects are consistent with those reported here.

FIGURE 1
Mean and Coefficient of Variation for Optical Patents as a Percentage of
Total Patents and Normalized Optical Words, 1982–2001^a



^a Total patents is the dependent variable, and normalized optical words is an independent variable.

figure raised the concern that some exogenous factor or omitted variable was driving both patenting and mentions of optics in the letters to shareholders. Yet the increasing coefficient of variation (standard deviation/mean, listed at the bottom of the diagram) suggested that firms did not converge toward a common level of intensity either in CEO attention to optics or in optical patenting patterns.

Table 1 reports the bivariate correlations of all measures. The three independent variables of interest—attention (normalized word counts), incentives (sales to carriers), and capabilities (optical product experience)—are all positively correlated with the dependent variable (optical patents as a percentage of total patents). The remaining variables are mainly correlated with the expected signs, though CEO demographic characteristics are not significantly associated with the dependent variable (but largely have the expected effects in regression analyses). Variance inflation factors testing for problems in collinearity did not indicate problems for the variables included in regressions. One might be concerned about the high correlation (.91) between the log of sales to carriers (the measure of incentives) and the log of sales (used as a control for scale); however, the variance inflation factors are each around 7 and therefore well below both the warning levels of 20 suggested by Greene (2003) and of 10 proposed by Chatterjee, Hadi, and Price (2000). Nevertheless, to alleviate the potential for worries about multicollinearity, I tested the regressions without the control for scale and also with a dummy variable to represent incentives (indicating

the presence of substantial sales to carriers in lieu of continuous carrier sales). Results were robust to these alternatives.

Main Effects: The Roles of Cognition, Incentives, and Capabilities

Despite the heterogeneity of the sample, in regression analyses I found systematic patterns of association between CEO attention to optical technologies and subsequent response to the fiber-optic revolution. Table 2 reports results for the test of the effect of attention on shaping investment in optical technologies as well as the related effects of organizational incentives and capabilities. The dependent variable is optical patenting emphasis (optical patents as a percentage of total patents), and the models are thus specified as two-sided Tobit equations (censored at 0 and 100 percent). The independent variables were lagged three years because it takes some time for organizations to respond to a change in top managers' emphasis on a technology. I also tested one-, two-, and four-year lag structures, obtaining similar results. I centered the main effect terms prior to creating the interaction terms to deal with the multicollinearity between main effects terms and interaction (Aiken & West, 1991), though I report uncentered statistics in Table 1.

Model 1 examines the effects of the controls, including the lagged dependent variable, scale (log of sales), resources (operating income as a percentage of sales), CEO demographic characteristics (technical background and job and firm tenure),

TABLE 1
Bivariate Correlations in Sample^a

Correlations	Minimum	Maximum	Mean	s.d.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1. Optical patents as a percentage of all patents	0.00	1.00	0.05	0.12																		
2. Normalized count of optics-related words	0.00	0.04	0.002	0.004	.41*																	
3. Sales to carriers ^b	0.00	11.24	6.23	2.60	.18*	-.10*																
4. Optical product experience	0	24	5.74	6.86	.31*	.06	.57*															
5. Sales ^b	0.48	11.28	6.98	2.74	.14*	-.15*	.91*	.61*														
6. R&D spending ^b	-6.91	8.73	4.48	2.85	.17*	-.13*	.88*	.53*	.93*													
7. Operating income ^c	-5.79	0.66	0.09	0.44	.04	.07	.24*	.07	.27*	.16*												
8. CEO technical background	0.00	1.00	0.62	0.49	-.01	.16*	-.15*	-.12*	-.21*	-.20*	-.07											
9. CEO job tenure	1	32	8.39	7.01	-.01	-.01	-.28*	-.32*	-.27*	-.23*	-.08	.18*										
10. CEO firm tenure	1	50	17.74	12.51	.10*	-.05	.21*	.29*	.36*	.33*	.00	.22*	.30*									
11. CEO outside experience	0	1	0.68	0.47	-.01	.01	-.26*	-.25*	-.27*	-.26*	.12*	-.18*	-.00	-.69*								
12. Focus on cellular technologies	0	1	0.41	0.49	-.07	-.19*	.28*	.05	.20*	.23*	-.12*	-.01	-.07	.01	-.21*							
13. Incumbent	0	1	0.73	0.44	.01	-.15*	.29*	.28*	.28*	.24*	-.10*	.13*	.02	.41*	-.33*	.22*						
14. Early entrant	0	1	0.17	0.37	-.16*	.04	-.20*	-.24*	-.21*	-.17*	.09*	-.06	-.03	-.33*	.23*	-.13*	-.75*					
15. Later entrant	0	1	0.10	0.30	.18*	.18*	-.18*	-.12*	-.15*	-.15*	.04	-.12*	.01	-.20*	.20*	-.16*	-.55*	-.15*				
16. Competitor patents/1,000	2.52	5.81	4.44	1.01	.04	.12*	.03	.16*	.01	.04	.14*	-.12*	-.18*	-.22	.08	-.04	-.32*	.16*	.27*			
17. Demand ^d	27.16	103.60	50.29	25.70	.09*	.08	.07	.18*	.04	.05	.02	-.13*	-.14*	-.15*	.05	.01	-.30*	.05	.37*	.63*		
18. Market trends ^e	0.57	1.30	0.43	0.38	.10*	.07	.07	.18*	.04	.04	.02	-.13*	-.15*	-.15*	-.05	.00	-.29*	.07	.34*	.56*	.94*	
19. Time trend	1985	2001	1994	4.61	.06	.10*	.06	.21*	.03	.05	.11*	-.17*	-.22*	-.23*	.08	-.02	-.36*	.16*	.33*	.86*	.85*	.82*

^a $n = 528$.

^b Logarithm.

^c As a percentage of sales.

^d From Value Line data on carriers' capital expenditures.

^e Count of *Wall St. Journal* articles mentioning optics.

TABLE 2
Results of Two-Sided Random-Effects Tobit Analyses for Relationship between Cognition, Capabilities, and Incentives, 1982–2001^a

Variable and Definition	1 Controls	2 Cognition	3 Incentives	4 Capabilities	5 Full Model	6 Interaction with Incentives	7 Interaction with Capabilities	8 Both Interactions
Cognition (Attention): Normalized optical words _{<i>t</i>−3}		13.452*** (2.342)			12.912*** (2.316)	13.455*** (2.421)	18.141*** (3.165)	18.622*** (3.151)
Incentives: Sales to carriers _{<i>t</i>−3}			0.012 ⁺ (0.06)		0.017** (0.006)	0.001 (0.007)	0.005 (0.008)	0.0001 (0.006)
Capabilities: Years optical product experience _{<i>t</i>−3}				0.006*** (0.002)	0.006*** (0.001)	0.000 (0.002)	0.001 (0.001)	−0.0002 (0.002)
Interactions: Cognition × incentives _{<i>t</i>−3} (Words × sales to carriers)						−3.609** (1.242)		−2.757* (1.287)
Cognition × capabilities _{<i>t</i>−3} (Words × years experience)							−1.140** (0.359)	−0.804* (0.349)
Lagged dependent variable: Optical patents as a percentage of total patents _{<i>t</i>−3}	0.149* (0.066)	0.130 ⁺ (0.070)	0.075 (0.070)	0.241*** (0.068)	0.130 ⁺ (0.067)	0.064 (0.072)	0.032 (0.072)	0.056 (0.070)
Scale: Log of sales _{<i>t</i>−3}	0.030*** (0.005)	0.043*** (0.006)	0.020** (0.007)	0.051*** (0.008)	0.034*** (0.009)	0.064*** (0.010)	0.043*** (0.009)	0.068*** (0.010)
Resources: Operating income percentage _{<i>t</i>−3}	0.278*** (0.067)	0.194** (0.065)	0.291*** (0.065)	0.228*** (0.070)	−0.306*** (0.065)	0.313*** (0.065)	0.212*** (0.065)	0.326*** (0.065)
CEO demographic characteristics: CEO technical background _{<i>t</i>−3}	0.050*** (0.015)	0.064*** (0.015)	0.043** (0.014)	0.063*** (0.018)	0.023 ⁺ (0.014)	0.036* (0.014)	0.055*** (0.015)	0.033* (0.014)
CEO job tenure _{<i>t</i>−3}	0.004*** (0.001)	0.001* (0.000)	0.005*** (0.001)	0.001 (0.001)	0.003** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
CEO firm tenure _{<i>t</i>−3}	−0.001 (0.001)	0.001 (0.000)	−0.001* (0.001)	0.001 ⁺ (0.001)	−0.001 ⁺ (0.001)	−0.001 (0.001)	−0.001 ⁺ (0.001)	−0.001 ⁺ (0.001)
Control for cellular focus: Presence in cellular technologies	−0.400*** (0.033)	0.049** (0.016)	−0.067*** (0.015)	0.064*** (0.019)	−0.113*** (0.020)	−0.102*** (0.019)	−0.007 (0.019)	−0.105*** (0.018)
Entry timing: Early entrant, 1986–91	−0.272*** (0.043)	−0.185*** (0.047)	0.037 (0.038)	−0.008 (0.055)	−0.190*** (0.040)	−0.138*** (0.039)	−0.209* (0.101)	−0.141*** (0.038)
Later entrant, 1992–2001	0.079* (0.039)	0.064 (0.047)	0.192*** (0.038)	0.228*** (0.045)	0.023 (0.039)	0.115* (0.056)	0.029 (0.045)	0.130*** (0.045)
Time trend: Optical words in WSJ/1,000 _{<i>t</i>−3}	0.030 (0.048)	−0.022 (0.045)	0.029 (0.045)	−0.208*** (0.056)	−0.223*** (0.051)	−0.092 ⁺ (0.050)	−0.077 (0.054)	−0.090 ⁺ (0.049)
Constant: Constant	−0.218*** (0.043)	−0.437*** (0.053)	−0.465*** (0.065)	−0.570*** (0.081)	−0.267*** (0.066)	−0.566*** (0.086)	−0.362*** (0.070)	−0.591*** (0.084)
Log-likelihood	116.637	130.579	109.540	115.917	138.615	143.791	137.334	146.458
Δdf		1 from baseline	1 from baseline	1 from baseline	3 from baseline	1 from model 5	1 from model 5	2 from model 5
Likelihood-ratio χ^2		27.88***	−14.19	−1.44	43.96***	10.35**	7.12*	15.69***

^a Variables were measured with three-year lags. The dependent variable was optical patenting emphasis (optical patents as a percentage of total patents). *n* = 528. Standard errors are in parentheses.

⁺ $p \leq .10$
* $p \leq .05$
** $p \leq .01$
*** $p \leq .001$

presence in cellular technologies, timing of entry, and a time trend. As expected, the significant parameter estimates indicate that optical patenting is greater for larger firms and those with slack resources and lower for those focused on cellular technologies. Of particular note, the coefficient on the early entrants dummy (in this and subsequent models) is largely negative (relative to incumbents, which is the omitted category), and that for later entrants is largely positive. This result suggests that the patenting rate in optics is associated with the timing of entry relative to the development of the technology. Those firms that entered after 1991, when some of the uncertainties about the potential impact of optical technologies were resolved, were more likely to invest in optical technologies.

Model 2 shows the independent effect of the measure of cognition (attention as measured by word counts in the letters to shareholders). Model 3 examines the role of incentives (serving carriers), and model 4 examines the role of capabilities (optical product experience). Model 5 includes all three explanations. Supporting Hypothesis 1, the measure of CEO attention performs as expected: changes in CEO attention to optics are significantly, positively associated with subsequent changes in optical patenting, even when organizational factors are accounted for. Note that the inclusion of measures of CEO demographic characteristics does not eliminate the effect of CEO attention, suggesting that these are separate constructs. Firms that served carriers, and therefore had an incentive to invest in optics, were also more likely to patent in optics at higher than average rates. This effect is only marginally significant when entered individually but increases in significance when entered along with cognition and capabilities (thus supporting Hypothesis 2). And increases in optical product experience are associated with subsequent increases in optical patenting (supporting Hypothesis 3). Likelihood tests suggest that only the models including CEO attention significantly improve the fit over the baseline ($p < .0001$). However, although adding incentives and capabilities individually does not improve the fit over the baseline (models 2 and 3), their inclusion with CEO attention in model 5 significantly improves the fit over a model with attention alone (model 4; likelihood-ratio $\chi^2 = 16.07$, $df = 2$, $p < .001$).

The results are statistically and economically significant. Using model 5, a one standard deviation increase in attention to optics is associated with a 5.8 percentage point absolute increase in optical patenting (at the mean of 4.6 percent, this is a 104 percent increase in optical patenting emphasis). One standard deviation increases in sales to carri-

ers and in optical product experience are associated with 96 and 89 percent increases in optical patenting, respectively. The results are robust to different lag structures, maintaining the positive association between words and patenting for one-, two-, and four-year lags (though significance is attenuated).

Relationships among Cognition, Incentives, and Capabilities

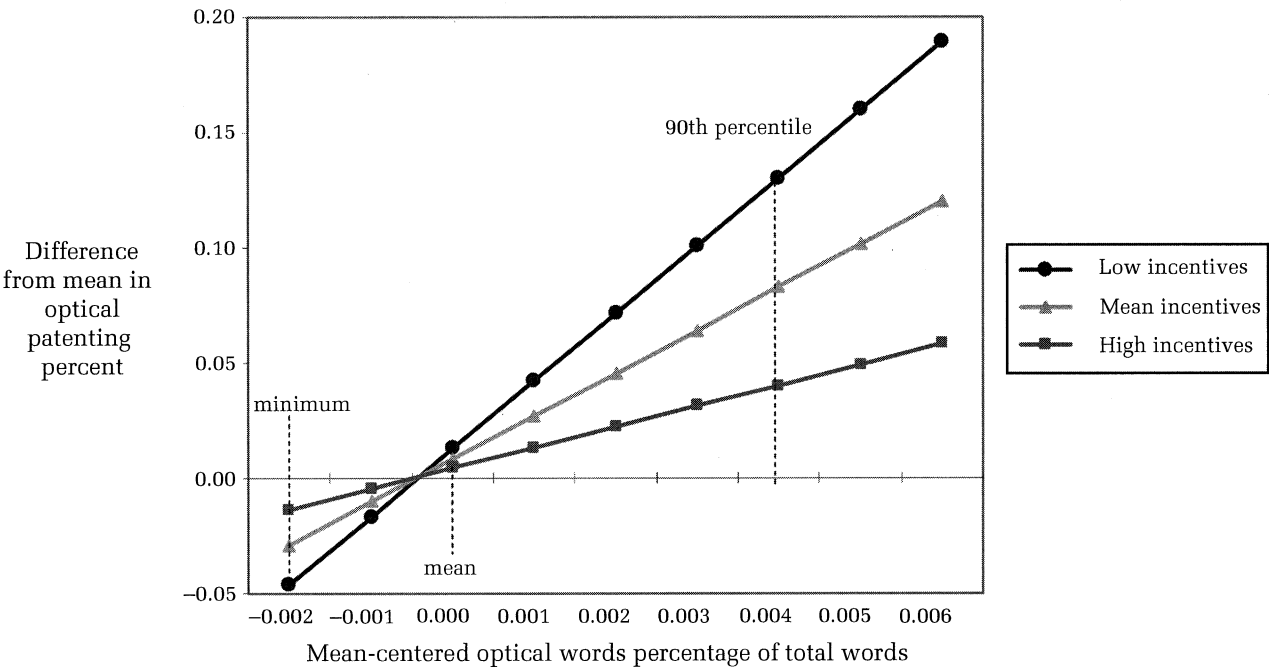
The next step in the analysis explores the relationships among CEO attention and organization-level incentives and capabilities in shaping firm response to optics. Model 6 reports the interactions between cognition and incentives, and model 7, that between cognition and capabilities. These effects are entered jointly in model 8. Each model improves the goodness of fit over model 5. The interactions of cognition with incentives and capabilities are both negative and significant, thus providing support for Hypotheses 4a and 5a. The effect of CEO attention to optics on subsequent technical investments is decreasing in incentives to invest and in years of optical product experience, though this does not take away from the overall positive effect of attention.

Figures 2a and 2b graph the interaction effects for incentives and capabilities, respectively, on the basis of model 8. These charts examine the incremental change from the mean in optical patenting based on changes in CEO attention to optics for low (10th percentile), mean, and high (90th percentile) levels of incentives or capabilities.⁹ The variables are all mean-centered so that a zero value represents the mean. These graphs show that where CEO attention to optics is very low, a firm's optical patenting rates tend to be lower than the sample mean. Increases in attention have a positive effect on subsequent optical patenting rates; however, both incentives to invest and prior related competence attenuate this effect. Changes in CEO attention have the least impact on firm technical investment when incentives or capabilities in the focal arena are high, but, in most other cases, the marginal effect of CEO attention remains quite substantial.

Stepping back, one can see that these results imply that changes in CEO attention to optics have an important main effect on changes in optical patenting rates and that this effect is intensified if firms do not have prior related compe-

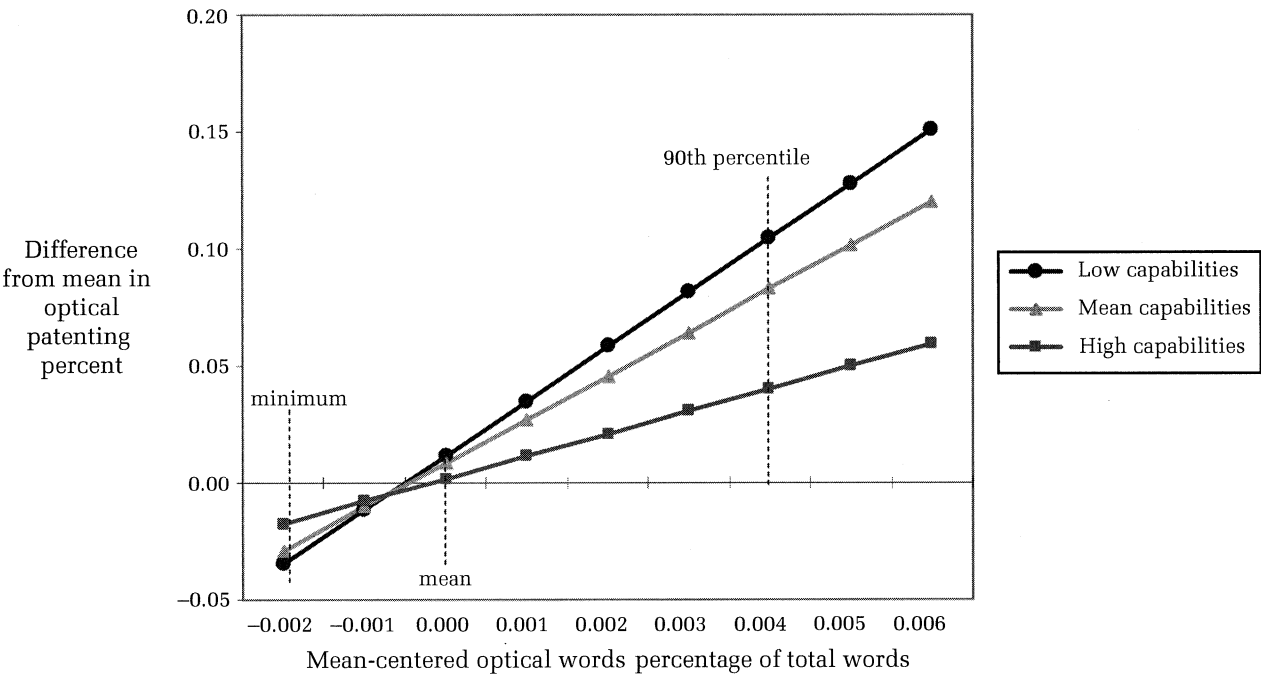
⁹ The tenth percentile of capabilities is 0 years of experience. This left truncation accounts for the relative closeness of the "minimum" line to the "mean" line.

FIGURE 2
Interaction between CEO Attention and Incentives, Mean-Centered^a



^a From model 8, all controls held constant, capabilities at the mean.

Interaction between CEO Attention and Capabilities, Mean-Centered^a



^a From model 8, all controls held constant, incentives at the mean.

tences in the optical arena or incentives to invest. CEOs can counteract a lack of these organization-level factors through increased attention to the

new technology. This effect is attenuated, however, as the incentives to invest or the years of experience increase.

Testing for Reverse Causality

One obvious alternative to the above interpretation of these findings on CEO attention is that the structure of causality is reversed: rather than attention leading to patenting, one might imagine that patenting led to subsequent CEO recognition of the importance of fiber optics, the focal technology here. Under this scenario, individual scientists or middle managers would make decisions to explore optical technologies and, if these investments proved to be successful, the outcome would be subsequent patents *and* mentions of optical technologies by their firm's CEO in the letter to shareholders. To assess the main direction of the effect, I tested cognition "in reverse" by regressing the measure of attention against the measures of response and found no significant effect (details are available from the author). This analysis showed no association between prior patenting activity and subsequent CEO attention to optical technologies, a finding that is at least consistent with the hypothesis that attention is not simply driven by an organization's prior technical investments. I did find, however, that a CEO background in technology does predict the level of mentions of optics in the shareholder letters, implying that attention may mediate this demographic characteristic. It may be useful, therefore, to think of demographic characteristics as a representation of a stable managerial capability, while attention to a particular issue should be viewed as a context-specific interpretation made by managers and influenced by multiple (often unmeasured) factors.

These results reinforce the view that a CEO's attention can shape subsequent investment patterns in a way that goes above and beyond the effects of organizational factors. On the other hand, the process by which a CEO develops a viewpoint about technology is likely highly complex. Although the effect is not significant in my analyses, it seems probable that prior technical activities have some influence on cognition in an iterative process of technical exploration and top management attention. Descriptive data from the companies studied as well as interviews with executives and engineers suggested that, because of the high uncertainty surrounding the focal technology, many types of exploration occurred within these firms. One could speculate that, given the information from these explorations, CEOs would make interpretations about the importance of particular technologies. These interpretations then legitimize a strategic direction, leading to both increased research efforts and patenting of prior research in that domain. As such, CEOs appear to play a critical

role in shaping firm response to technical change by taking a stance in an uncertain world, focusing attention on a particular technology, and legitimizing one course of action among many options.

DISCUSSION

The purpose of this study of communications technology firms facing the fiber-optic revolution was to understand the extent to which CEO attention could help explain heterogeneous firm response to technical change. This analysis built on the managerial cognition and management of technology literatures and focused on the effect of changes in attention on subsequent changes in investment patterns under highly uncertain conditions. The results confirmed that changes in CEO attention to optics do affect the subsequent degree of optical patenting in a firm, even when these changes were considered alongside the effects of organizational incentives and capabilities. Indeed, the analysis showed that cognition, capabilities, and incentives were related in important ways in shaping firm response to change, for both incumbents and entrants into the new technical arena.

The approach in this article demonstrates one empirical methodology for exploring Ocasio's (1997) attention-based theory. Although these analyses lend weight to the concept that top managers play a crucial role in both interpreting the external environment and shaping the internal response to it, the approach has limitations. First, the measure of attention used here is admittedly quite abstracted from the actual interpretive processes of CEOs. But for a researcher constrained by the need to obtain comparable quantitative measures of attention across firms and over time, a count of words in letters to shareholders may be the best measure available.

Second, the process by which CEOs develop their interpretations of new technologies is likely dynamic and iterative. Although quantitative statistical analyses of managerial cognition are appealing in that they can establish systematic relationships, they are limited in their ability to capture the underlying mechanisms that connect cognition with strategic action and to explain the process by which cognition is interrelated with incentives and capabilities. Attention as measured by counts of words in a letter to shareholders is a "snapshot" that may best be understood as the adoption and legitimization of a particular set of interpretations by CEOs (Oliver & Johnston, 2000). As others have argued, we should not reify "the organization" as a cognizer but rather understand how a collective viewpoint and ultimately a collective decision

might emerge through actors' interactions (Garud & Karnøe, 2001; Spender, 1998; Weick & Roberts, 1993). In further research, particularly qualitative studies of managers' practices, scholars could seek to unpack these mechanisms in greater detail (cf. Kaplan, *in press*).

Despite these limitations, the findings from this study shed light on the ways in which CEO attention can matter in such highly turbulent contexts as radical technical change. First, I found a systematic association between changes in CEO attention and subsequent changes in investment patterns. Significant theory and evidence—related to both economic arguments about incentives and evolutionary views on capabilities—would suggest that managerial cognition should not matter in the context of technical change. Instead, even when accounting for such organizational factors, I found that changes in a CEO's attention to a new technology were associated with subsequent changes in investment in the new arena. The test of reverse causality further suggested that this effect was not simply the product of a rising industry tide. When a firm is faced with an uncertain future, some foresight seems to be useful for adaptation (Ahuja, Coff, & Lee, 2005; Henderson, 2000). The findings are also consistent with research suggesting that changes to a top management team can affect firm response to technical change (Tushman & Rosenkopf, 1996; Virany et al., 1992). Showing that context-specific managerial cognition affects the strategic direction of a firm provides an additional mechanism to explain why such management turnover might be needed. If managers cannot change their minds (for example, if they do not focus attention on emerging technologies), then these managers will need to be replaced if their organizations are to adapt.

Second, these results also offer some sense of the relationship between managers' demographic characteristics and their ongoing interpretations of their environments. Researchers know, for example, that managers' occupational orientations can shape the "thought worlds" they bring to their work (Carlile, 2002; Dougherty, 1992). Using this logic, it may be useful to think of managers as accumulating knowledge through a wide variety of experiences, including their education, their jobs, and their functional experience. These accumulations can be captured in one way or another by demographic measures. In any particular situation, managers draw on their repertoires of knowledge accumulations to make a context-specific interpretation of the situation (Bogner & Barr, 2000; Elsbach, Barr, & Hargadon, 2005; Lant, 2002) in ways that may be conditioned by other factors, such as social interaction (Chatto-

padhyay, Glick, Miller, & Huber, 1999). The key idea (which I captured through a direct measure of attention) is that managers' interpretations are situated, transitory, and temporally bounded, while demographic measures capture the more stable repertoire of knowledge that managers bring to each situation.

Third, this article contributes to recent scholarship attempting to tease out the differential effects of managers and organizational factors such as incentives (Johnson & Hoopes, 2003) and capabilities (King & Tucci, 2002). The results suggest that cognition is tightly intertwined with organizational incentives and capabilities in shaping firm response to technical change. In both cases, it appears that CEO attention to a new technology has the highest impact on strategic action when organizational factors are not aligned with the opportunity. The incremental effect of attention is greatest when organizational reorientations are required.

Cognition interacts with organizational incentives to invest. In line with economic arguments, incentives alone can lead to differences in performance: presence in a context that provides strong incentives for investment in a particular technological innovation has an important main effect on increasing levels of resource allocation. This finding is consistent with Burgelman's (1994) story about Intel's exit from memory chips. The autonomous decision-making approach led the organization into markets that pulled Intel toward microprocessors and away from memory (DRAMs). In a situation such as this, where incentives to invest are quite high, the incremental effect of CEO attention to a new technical arena may be minimal. But when incentives to act are low, cognition will have an increasingly substantial effect on outcomes.

Cognition also interacts with the capabilities of an organization. The findings suggest that if prior competence is related to a new field, then the direct effect of managerial cognition is attenuated; and, conversely, when relevant capabilities are not present, the effect of cognition on action increases substantially. These results are consistent with case study research suggesting that CEO attention can compensate for a lack of capabilities (Rosenbloom, 2000). A CEO's attention to a new technology might usefully be thought of as a forward-looking view that guides search efforts in new arenas (Gavetti & Levinthal, 2000). Where relevant capabilities already exist, CEO cognition may only serve to modify ongoing activities in an organization. Having such "preadaptive" capabilities may improve a firm's chance for responding to environmental change (Cattani, 2005; Helfat, 1997). But these capabilities may not be enough. As Tripsas and

Gavetti (2000) demonstrated, even when a firm has the needed technical capabilities, it may be unable to respond effectively to a new technology if managers' beliefs are not correctly aligned with the opportunity. As such, the research reported here sheds further light on the suggestion that managers can moderate the effect of firm-level experience. By measuring managerial cognition directly, I was able to tease apart the separate but related effects of cognition and organizational experience.

The implication of this work is that it is useful to take into account the joint effect of cognition, capabilities, and incentives in seeking to understand the potential for strategic reorientation in the face of change. Each of these three factors is separately important in shaping outcomes, and their alignment toward the adoption of a new technology leads to the greatest change. On the other hand, manager-level cognition can compensate when organization-level factors are not in place, and therefore it could be considered a dynamic *managerial* capability helping firms to adapt to changes in their environments (Adner & Helfat, 2003). CEO attention to a new opportunity can be the deciding factor in reorienting an organization to take advantage of it.

From a normative standpoint, these findings imply that firms that need to change face two problems. One is that managers often find it hard to change their views, particularly if such views are derived from highly tacit accumulated knowledge that underpins the activities of the firm. The other is that change may require a break in the links of a deeply embedded organizational architecture that ties together cognition, capabilities, and incentives (Henderson & Clark, 1990; Kaplan & Henderson, 2005). Established firms, particularly successful ones, have developed smoothly functioning systems of routines (Jacobides & Winter, 2005), not just for production, but also for the ways managers understand markets and technologies and the kinds of incentives that are meaningful. Although extremely useful in stable contexts, once a firm is faced with a radical technical change, such a system can inhibit adaptation. Therefore, change requires these links to be broken, and the evidence from this study suggests that it is a firm's CEO (or other top managers) who must break the links by shifting attention to the new technology.

In summary, this study contributes to understanding of how firms respond to technical change. The cognitive perspective proposed here suggests that the degree to which managers' cognition changes may affect the degree of a firm's inertia, but that these effects depend crucially on other elements of the organization's architecture: organiza-

tional incentives and capabilities. The forces for inertia and adaptation may both be present within a firm, and managerial cognition can be a critical arbiter of the degree and direction of firm action.

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Sarah Kaplan (slkaplan@wharton.upenn.edu) is an assistant professor of management at the University of Pennsylvania's Wharton School. Her research seeks to unpack the role of cognitive framing in shaping the evolution of technologies, heterogeneous organizational response to discontinuities, and strategy making inside organizations. She received her Ph.D. from MIT's Sloan School of Management.



This article continues with an appendix.

APPENDIX Sample Description

TABLE A1
Stratified Random Sample of 71 Firms in the Communications Technology Industry

Entry	Average Size			
	Large	Medium	Small	Total
Incumbent, pre-1985	<i>Total population: 20</i> <i>Number of firms sampled: 19</i> <i>Sampling weight: 95%^a</i> 3Com Corp. Alcatel AT&T Corp. Corning Inc. Ericsson (L M) Tel Fujitsu Ltd. General Instrument Corp. Harris Corp. Hubbell Inc. ITT Industries Inc. Kyocera Corp. Motorola Inc. National Semiconductor NEC Corp. Nokia Corp. Nortel Networks Corp. Raychem Corp. Siemens AG Texas Instruments Inc.	<i>Total population: 66</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 8%</i> Comverse Technology ECI Telecommunications General Datacomm Inds. Paradyne Corp. Triangle Industries Inc.	<i>Total population: 116</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 4%</i> Andrea Electronics Corp. Interdigital Communications Microwave Filter Co. Inc. Porta Systems Corp. Telebyte Inc.	<i>Total population: 202</i> <i>Number of firms sampled: 29</i> <i>Sampling weight: 14%</i>
Early entrant, 1986–91	<i>Total population: 4</i> <i>Number of firms sampled: 4</i> <i>Sampling weight: 100%</i> Bay Networks Inc. Cisco Systems Inc. Qualcomm Inc. Reliance Electric Co.	<i>Total population: 25</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 20%</i> Atmel Corp. DMC Stratex Networks Maxim Integrated Newbridge Networks Vitesse Semiconductor	<i>Total population: 59</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 8%</i> California Micro Devices Franklin Telecom. Microtel International Openroute Networks Optek Technology Inc.	<i>Total population: 88</i> <i>Number of firms sampled: 14</i> <i>Sampling weight: 16%</i>
Mid entrant, 1992–96	<i>Total population: 3</i> <i>Number of firms sampled: 3</i> <i>Sampling weight: 100%</i> Lucent Technologies Inc. STMicroelectronics NV Superior Telecom Inc.	<i>Total population: 53</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 9%</i> Ciena Corp. JDS Uniphase Corp. Madge Networks NV Remec Inc. Stratacom Inc.	<i>Total population: 130</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 4%</i> Anadigics Inc. Brooktrout Inc. Fibercore Inc. Optical Cable Corp. SBS Technologies Inc.	<i>Total population: 186</i> <i>Number of firms sampled: 13</i> <i>Sampling weight: 7%</i>
Bubble entrant, 1997–2001	<i>Total population: 13</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 38%</i> Agilent Technologies Inc. Conexant Systems Inc. Epcos AG Infineon Technologies AG ON Semiconductor	<i>Total population: 23</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 22%</i> Brocade Communications Corvis Corp. Intersil Corp. Research in Motion Terayon Communication Systems	<i>Total population: 105</i> <i>Number of firms sampled: 5</i> <i>Sampling weight: 5%</i> Bookham Technology Finisar Corp. MCK Communications Metalink Ltd. Turnstone Systems Inc.	<i>Total population: 141</i> <i>Number of firms sampled: 15</i> <i>Sampling weight: 11%</i>
Total	<i>Total population: 40</i> <i>Number of firms sampled: 31</i> <i>Sampling weight: 78%</i>	<i>Total population: 167</i> <i>Number of firms sampled: 20</i> <i>Sampling weight: 12%</i>	<i>Total population: 410</i> <i>Number of firms sampled: 20</i> <i>Sampling weight: 5%</i>	<i>Total population: 617</i> <i>Number of firms sampled: 71</i> <i>Sampling weight: 12%</i>

^a Perkin Elmer was eliminated because of problems in matching data, owing to a series of ownership changes and spin-offs.